Mei Stoesz-Gouthro
V01004629
GIS Analysis
Lab 4

Results

Air quality measurements ($PM_{2.5}$) across China were interpolated using inverse distance weighted with cell sizes of 10 and 20 kilometers. Tables 1 and 2 summarize measurement statistics for the inverse distance weighted raster surfaces. The 10-kilometer resolution prediction error statistics analysis yielded a minimum of -9.43 $PM_{2.5}$, a maximum of 10.41 $PM_{2.5}$, and a range of 19.83 $PM_{2.5}$. The calculated mean and median values are -0.07 and 0.04 $PM_{2.5}$, respectively. The shape of the predicted error data is a bell curve, with the data with the highest frequency, residing in the middle of the range. Prediction errors were overlaid at monitoring stations across China. The prediction error point values are divided into six classes by standard deviation. Blue points represent underprediction and red represents an overprediction (see figures 1 and 3). The surface with a 10-kilometer raster (figure 1) displays overprediction points in the Eastern central-latitude region of China. Underprediction points are dispersed, with high values in the North. The 20-kilometer resolution prediction error statistics analysis yielded a minimum of -10.64 $PM_{2.5}$, a maximum of 7.75 $PM_{2.5}$, and a range of 18.39 $PM_{2.5}$. The calculated mean and median values are -0.15 and 0.06 $PM_{2.5}$, respectively. The shape of the predicted error data is a left-skew bell curve. The 20-kilometer raster (figure 3) displays dispersed overprediction points in the Eastern half of China. Underprediction points are dispersed, with high values forming a diagonal line from the Yunnan province (South-Western region) to the Liaoning province (North-Eastern). Although six classes were requested, the 20-km raster surface naturally resulted in only four classes.

Cumulative effects of sampled air pollution from 2016 to 2020 are displayed in figure 5. Provincial cumulative effects can be seen in figure 6. The provincial values were calculated by summing $PM_{2.5}$ values within provincial boundaries and values were then normalized by provincial land area (µg/m³ per km²) to account for size differences among provinces. Five classes were derived using natural breaks to maximize variability between groups, while minimizing it within groups. High pollution values are found in Xinjian province (North-West) and the central-Western area of the country (see figure 6). Low cumulative effects are found in the mountainous South-Western region and in central China. The provinces with the highest concentration of $PM_{2.5}$ are Henan (2.734), Tianjin (2.533), Shandong (2.51), and Hebei (2.477). The highest range of $PM_{2.5}$ values belong to Xinjian (418.749), Hebei (236.606), Sichuan (217.324), and Xizang (198.948). Cumulative effects variance is displayed in figure 7. The highest variance values are found in Xinjian and mid-level values in Hebei. The remaining provinces have relatively low $PM_{2.5}$ cumulative effect variance.

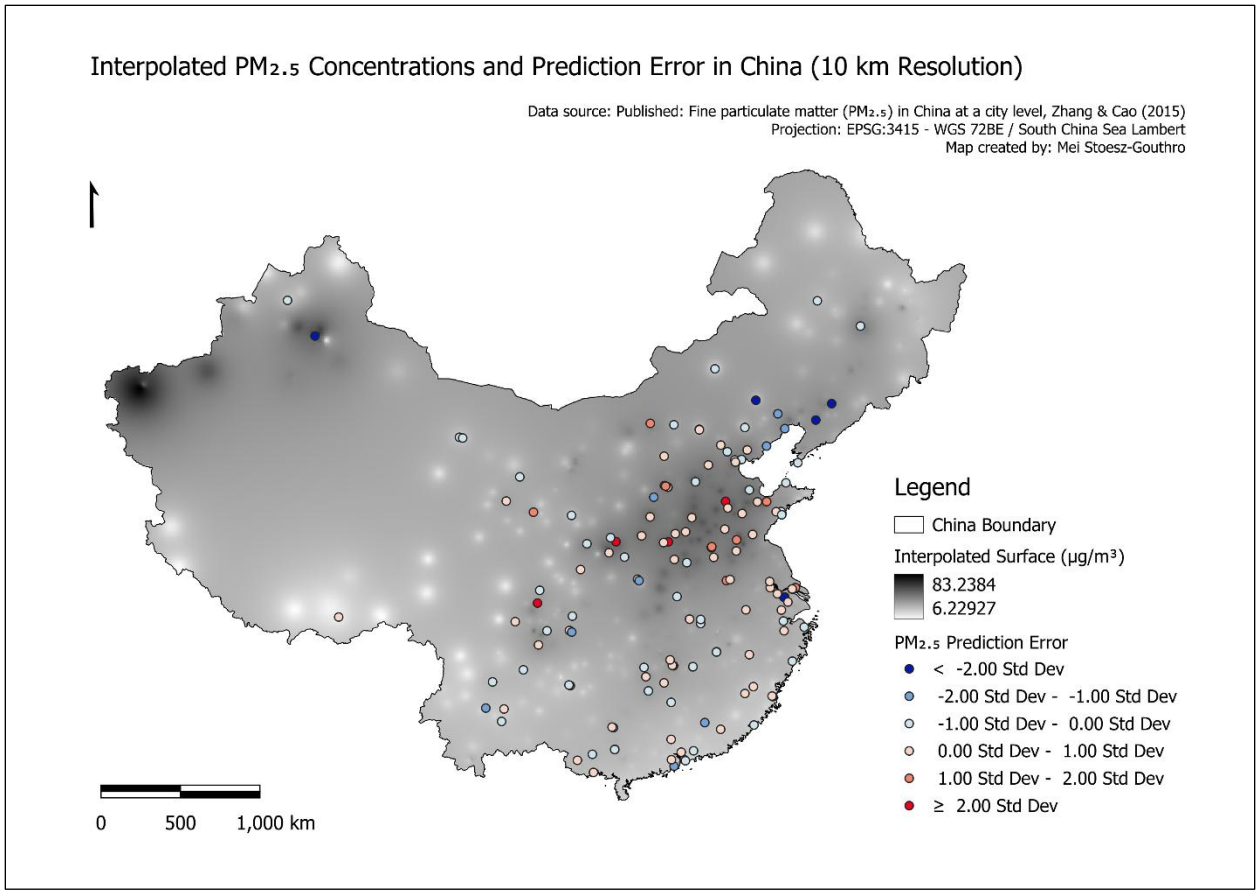

Figure 1. Inverse distance weighted (IDW) interpolation of $PM_{2.5}$ concentrations in China at 10 km resolution with overlaid prediction errors at monitoring stations across the China. Points show prediction errors classified by standard deviation, where blue indicates underprediction and red indicates overprediction.

Table 1. Interpolated PM2.5 Concentration Prediction Error Statistics (10 km Resolution)

Count	145
Minimum	-9.425
Maximum	10.409
Range	19.834
Sum	-9.582000
Mean	-0.066083
Median	0.041000
Standard Deviation	2.623511459734

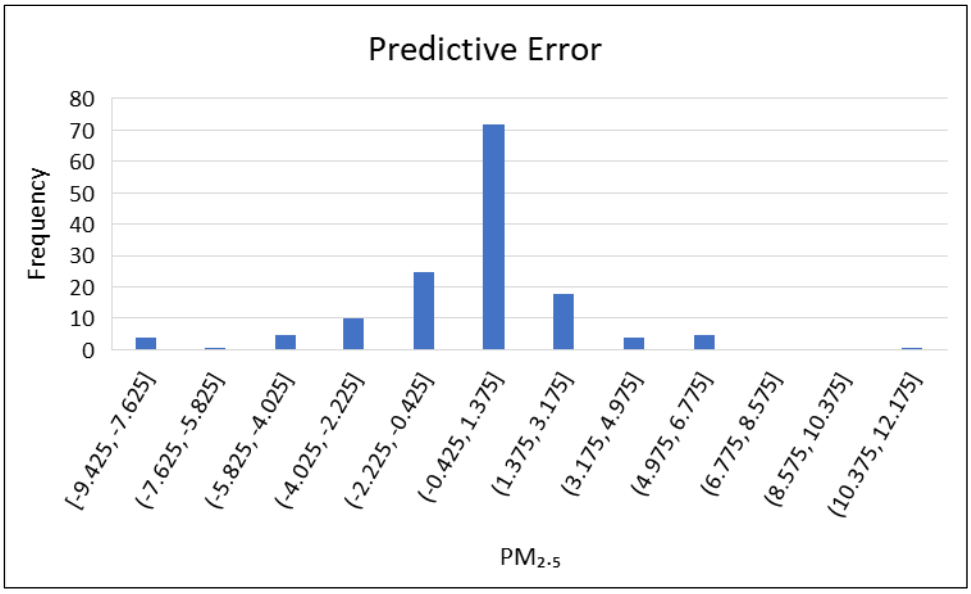

Figure 2. Interpolated PM2.5 concentration prediction error distribution (μg/m³) across the interpolated surface for China, at a 10 km resolution.

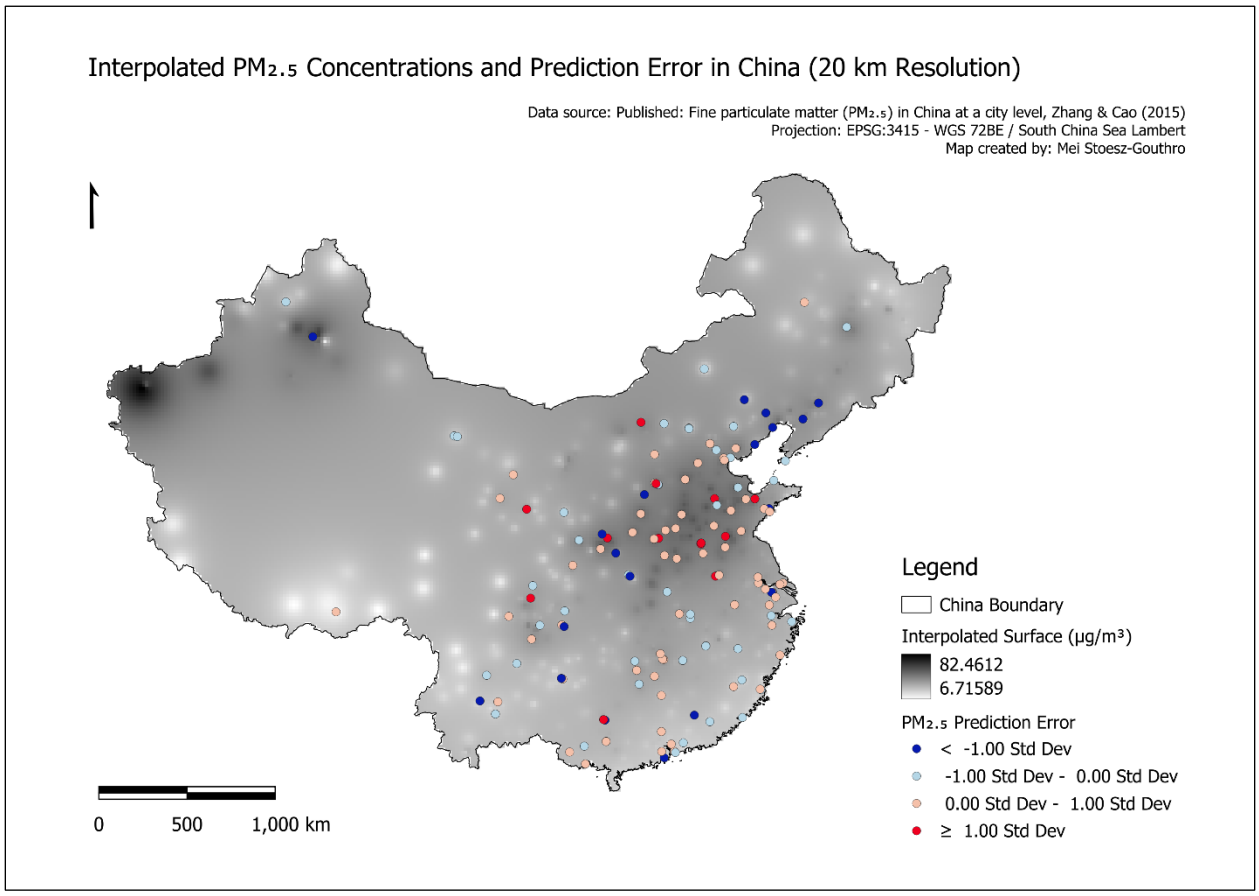

Figure 3. Inverse distance weighted (IDW) interpolation of $PM_{2.5}$ concentrations in China at 20 km resolution with overlaid prediction errors at monitoring stations across the China. Points show prediction errors classified by standard deviation, where blue indicates underprediction and red indicates overprediction.

Table 2. Interpolated PM2.5 Concentration Prediction Error Statistics (20 km Resolution)

Count	145
Minimum	-10.641
Maximum	7.751
Range	18.392
Sum	-21.175000
Mean	-0.146034
Median	0.060000
Standard Deviation	2.769288182943

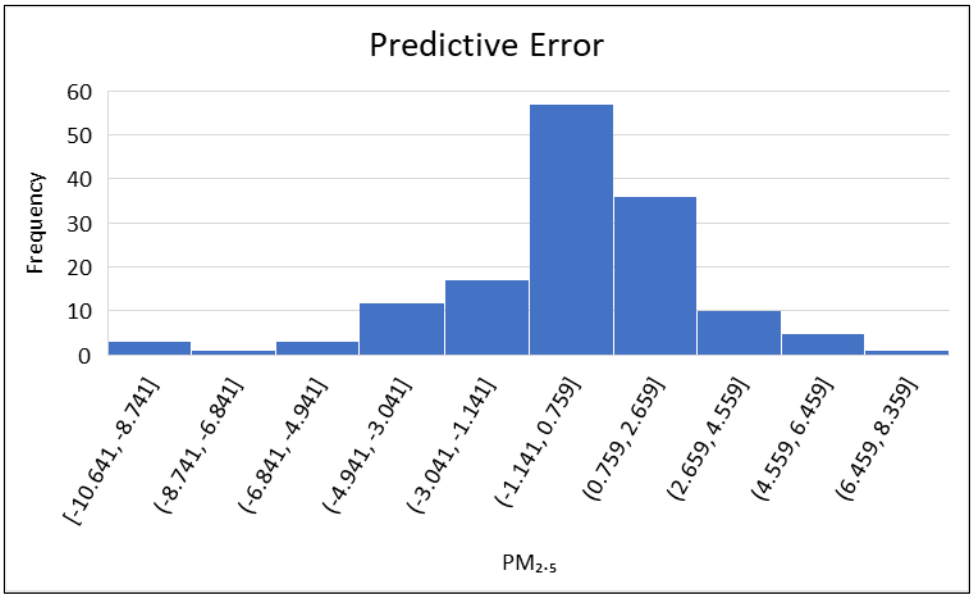

Figure 4. Interpolated PM2.5 concentration prediction error distribution (μg/m³) across the interpolated surface for China, at a 20 km resolution.

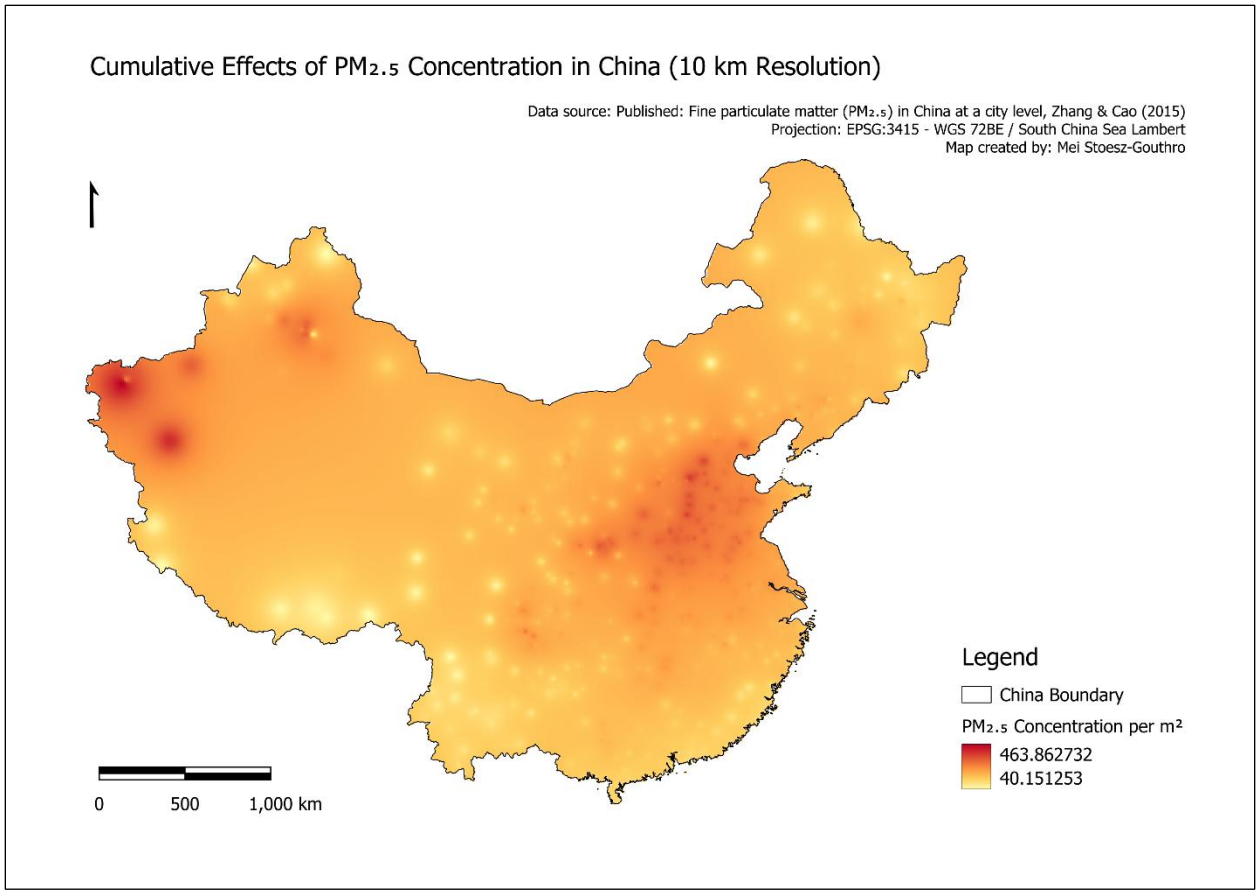

Figure 5. Cumulative effects of annual mean $PM_{2.5}$ concentrations across China at 10 km resolution. The surface was derived by summing five years of data and interpolated using inverse distance weighting (IDW). High values highlight regions experiencing the greatest multi-year particulate exposure.

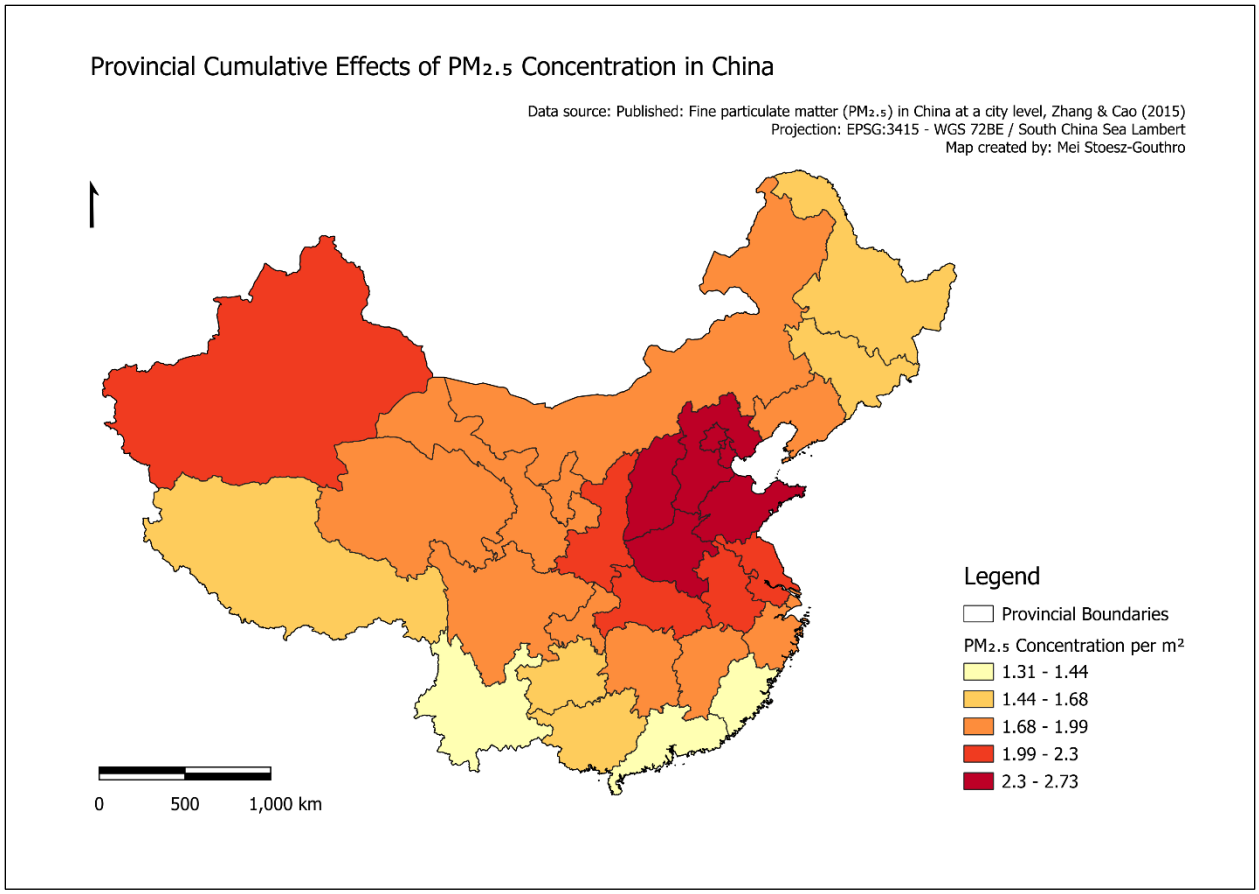

Figure 6. Provincial cumulative effects of annual mean $PM_{2.5}$ concentrations across China at 10 km resolution. The cumulative values were calculated for each province by summing $PM_{2.5}$ values within provincial boundaries. Values were normalized by provincial land area (μg/m³ per km²) to account for size differences among provinces. Class valued were calculated with natural breaks and highlight high $PM_{2.5}$ densities in Central East China and North West China.

Table 3. Provincial Cumulative Effects Statistics of $PM_{2.5}$ Concentration in China

Province	Minimum	Maximum	Range	Sum	Mean
Anhui	190.7306	285.902	95.17143	333250.2	229.5112
Beijing	208.0551	264.1445	56.0894	44151.24	238.6554
Chongqing	158.0183	216.7285	58.71013	167435.8	197.6809
Fujian	92.4757	178.1902	85.71445	174767.3	145.0351
Gansu	108.7329	238.9246	130.1917	796888.4	178.9554
Guangdong	99.42099	169.3125	69.89156	245887.4	142.8747
Guangxi	123.7918	186.3861	62.5943	379143.1	160.3142
Guizhou	88.58232	198.4642	109.8819	279804.6	157.7252
Hebei	132.6348	369.2412	236.6064	522182.3	246.778
Heilongjiang	60.02667	203.0703	143.0436	887066.8	153.6579
Henan	225.7605	363.8299	138.0694	477514.2	273.0213
Hubei	155.1853	291.6729	136.4876	424464.4	221.6524
Hunan	136.2342	237.5091	101.2749	408104	190.9705
Inner Mongol	54.18151	211.5367	157.3551	2469130	181.8478
Jiangsu	178.3612	306.4381	128.0768	232747.5	222.512
Jiangxi	132.3224	224.9649	92.64243	312179.7	182.9892
Jilin	98.30557	187.5738	89.26826	379500	168.7416
Liaoning	140.0759	217.1241	77.04825	313685.3	188.3996
Ningxia	146.3694	206.6542	60.28484	104959.6	185.441
Qinghai	54.04765	194.1893	140.1416	1338104	174.7328
Shaanxi	147.3581	332.8827	185.5246	493432.2	225.0033
Shandong	128.4969	315.131	186.634	420732.5	253.1483
Shanghai	172.4277	200.3712	27.9435	12326.59	186.7666
Shanxi	163.2354	314.5859	151.3505	419271.5	245.907
Sichuan	63.53917	280.863	217.3238	868273.4	174.3521
Tianjin	239.6392	288.1577	48.51857	32301.09	260.4926
Xinjiang	45.11425	463.8627	418.7485	4100917	221.2526
Xizang	40.15125	239.0991	198.9479	1800600	154.6509
Yunnan	53.69856	210.1102	156.4116	505994.9	131.9413
Zhejiang	100.8382	202.7981	101.9599	178782.1	176.3137

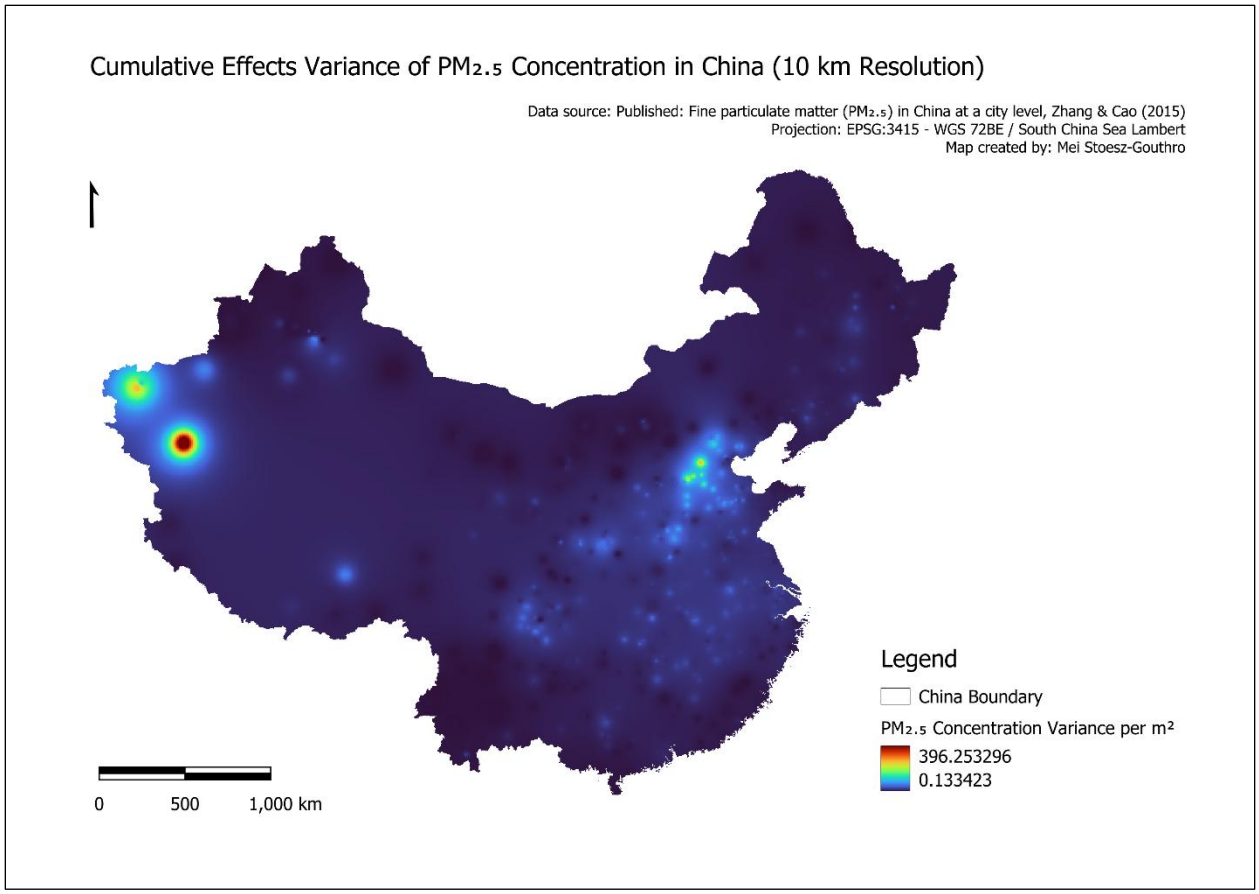

Figure 7. Spatial distribution of variance in modelled $PM_{2.5}$ concentrations across China (2016–2020) at 5 km resolution. Variance was calculated using statistics on annual IDW interpolated cells. Areas with higher variance indicate greater interannual instability in $PM_{2.5}$ concentrations, while lower variance reflects more stable conditions over time. Elevated variability is observed in parts of North West and Central East China.

Discussion

GIS software is utilized to create visual diagrams of $PM_{2.5}$ concentration across China. Multiple datasets can be used in analysis with endless combinations. Statistics can be performed with attribute tables from datasets. While data in table or chart format can be extremely insightful, visualization is excellent for analyzing spatial trends and dependence for the study topic. Science communication heavily relies on visuals to convey information to the public. GIS software can produce precise and accurate representations of datasets but dataset quality matters (Jerrett et al., 2005). Interpolation is directly influenced by the number of sample sites and their distribution. Ideally, a high number of sample sites and random or ordered dispersion is best for interpolation accuracy over spatial areas, though, sample sites are often spatially dependent (Lu & Wong, 2008). In this case, air quality reading sample sites are biased to the Eastern side of the country.

Inverse distance weighting is a simple yet effective technique for interpolating data onto a surface. This method places higher influence on nearby points and less influence on far points. Air quality near Heibei's border will be higher than $PM_{2.5}$ readings further from the province. Study areas with high numbers of

sample sites and a uniform distribution will produce higher accuracy interpolation data than areas with few or clustered sites. Low numbers of sampling sites produce interpolation surfaces with unreliable assumptions, while clustered sites can overrepresent areas (Lu & Wong, 2008). Large cells, like the 20-kilometer interpolation (figure 3) produce a smooth surface but can lose detail at the fine scale, especially in small, specific areas. Smaller resolutions, such as the 10-kilometer model (figure 1), preserve spatial detail in areas with adequate sample site numbers and distribution but can result in inaccurate interpolation in regions with little sites. Voronoi/Thiessen polygons are best suited for categorical data or data where polygons with abrupt boundaries make sense, such as precipitation measurements. Air quality interpolation with polygons would result in a highly inaccurate interpolated surface, as air moves freely around the Earth. Kriging interpolation is more computational than IDW but it provides best linear unbiased predictions (BLUP) based on spatial statistics. Spatial patterns in areas with few or no sites can be represented in the interpolated surface. Spatial trends must be known about the study area to produce an accurate Kriging interpolation model.

Decision making based on this IDW interpolation study provides a generalized representation of air quality throughout China. There are gaps of training sites in areas with low populations and harsh terrain or biomes. Kriging interpolation at a fine spatial resolution would likely produce more accurate results than the IDW analysis but requires more information on spatial autocorrelation. The data produced from this study is best suited for decision making in the Eastern parts of China. Integrating social and socioeconomic datasets into the decision would provide a holistic view for informed policy and decision making (Jerrett et al., 2005).

References

Jerrett, M., Arain, A., Kanaroglou, P., Beckerman, B., Potoglou, D., Sahsuvaroglu, T., Morrison, J., & Giovis, C. (2005). A review and evaluation of intraurban air pollution exposure models. *Environmental Health Perspectives, 113*(7), 889–897.

Lu, G. Y., & Wong, D. W. (2008). An adaptive inverse-distance weighting spatial interpolation technique. *International Journal of Geographical Information Science, 22*(9), 1049–1067.